

SQL Analysis Results

Revenue Leakage Investigation
FreshMart Retail Pvt. Ltd.

A concise summary of the executed SQL analysis and the results used to investigate revenue, stockouts, demand gaps and inventory performance.

Analysis area	Purpose
Revenue	Understand sales performance across regions and stores
Stockouts	Identify where product availability problems are concentrated
Demand gap	Find where forecast demand is higher than units sold
Inventory	Review utilization and overstock patterns
Rankings	Identify stores and products needing attention
Stockout impact	Examine revenue and demand gap during stockout records

Source: SQL queries and executed result screenshots supplied for this project. Values in this report are taken from those results.

1. Revenue Performance

Q1 — Total Revenue

How much total revenue is present in the dataset?

Metric	Result
Total Revenue	494,971,391.76 currency units

What this shows: The dataset contains about 494.97M currency units in recorded revenue.

Q2 — Revenue by Region

Which regions generate the highest revenue?

Region	Total Revenue
East	124,792,224.91 currency units
South	124,567,392.95 currency units
North	123,543,666.06 currency units
West	122,068,508.54 currency units

What this shows: East has the highest revenue, although the four regions are relatively close.

Q3 — Revenue by Store

Which stores generate the highest revenue?

Store	Total Revenue
S005	100,503,643.04 currency units
S003	100,034,405.71 currency units
S002	98,950,007.22 currency units
S004	98,057,144.10 currency units
S001	97,426,191.69 currency units

What this shows: S005 ranks first in revenue, while S001 ranks fifth among the five stores.

2. Stockout Analysis

Q4 — Stockouts by Region

Which regions have the highest number of stockouts?

Region	Stockout Count
West	673
North	661
East	647
South	604

What this shows: West has the highest stockout count. Stockouts occur across all four regions.

Q5 — Stockouts by Store

Which stores have the highest number of stockouts?

Store	Stockout Count
S004	547
S005	527
S002	523
S003	506
S001	482

What this shows: S004 has the highest count, followed by S005 and S002.

Q6 — Stockouts by Category

Which product categories are most affected by stockouts?

Category	Stockout Count
Groceries	535
Furniture	524
Clothing	510
Electronics	508
Toys	508

What this shows: Groceries has the highest category count, but the differences are small, showing a broad problem.

3. Demand Gap Analysis

Q7 — Demand Gap by Region

Which regions have the highest total demand gap?

Region	Total Demand Gap
South	94,132.02
West	93,166.38
North	92,201.80
East	90,667.72

What this shows: South has the highest total demand gap, with West close behind.

Q8 — Demand Gap by Store

Which stores have the highest total demand gap?

Store	Total Demand Gap
S005	75,416.22
S004	73,853.01
S002	73,837.66
S001	73,696.26
S003	73,364.78

What this shows: S005 has the highest store-level demand gap, while the overall values are fairly close.

Q9 — Demand Gap by Category

Which categories have the highest unmet demand?

Category	Total Demand Gap
Toys	74,913.95
Furniture	74,834.06
Clothing	74,534.04
Electronics	73,028.52
Groceries	72,857.35

What this shows: Toys has the highest category demand gap, closely followed by Furniture and Clothing.

4. Inventory Performance

Q10 – Inventory Utilization by Region

Which regions have the highest average inventory utilization?

Region	Average Utilization
West	49.90
South	49.84
East	49.82
North	49.64

What this shows: Utilization is very similar across regions, with West slightly highest.

Q11 – Lowest Inventory Utilization by Store

Which stores have the lowest average inventory utilization?

Store	Average Utilization
S001	49.55
S002	49.73
S005	49.81
S003	49.95
S004	49.96

What this shows: S001 has the lowest average inventory utilization.

Q12 – Overstock by Category

Which categories have the most overstock records?

Category	Overstock Count
Furniture	14,173
Toys	14,135
Clothing	14,116
Groceries	14,075
Electronics	14,013

What this shows: Furniture has the highest overstock count, although all categories are close.

5. Product & Benchmark Analysis

Q13 — Stores Above Average Revenue

Which stores generate above-average revenue?

Store	Total Revenue
S005	100,503,643.04 currency units
S003	100,034,405.71 currency units

What this shows: S005 and S003 are above the average store-level revenue calculated across the five stores.

Q14 — Products Below Average Utilization

Which products have below-average inventory utilization?

Product	Average Utilization
P0008	49.06
P0003	49.12
P0017	49.19
P0002	49.45
P0012	49.51
P0007	49.58
P0009	49.68
P0018	49.72
P0014	49.72
P0001	49.78

What this shows: P0008 has the lowest utilization among the products returned by this query.

6. Product & Store Rankings

Q15 / Q17 — Highest Product Demand Gaps

Which products have above-average demand gaps, and which are the top 10?

Rank	Product	Total Demand Gap
1	P0018	19,427.34
2	P0013	19,178.61
3	P0003	19,015.64
4	P0001	18,989.23
5	P0004	18,971.79
6	P0015	18,907.56
7	P0017	18,666.72
8	P0007	18,612.24
9	P0009	18,570.12
10	P0012	18,547.05

What this shows: P0018 has the highest total demand gap among the products shown.

Q16 — Store Revenue Ranking

Which stores rank highest by total revenue?

Rank	Store	Total Revenue
1	S005	100,503,643.04 currency units
2	S003	100,034,405.71 currency units
3	S002	98,950,007.22 currency units
4	S004	98,057,144.10 currency units
5	S001	97,426,191.69 currency units

What this shows: S005 ranks first, followed by S003.

Q18 — Store Stockout Ranking

Which stores have the highest number of stockouts?

Rank	Store	Stockout Count
1	S004	547
2	S005	527
3	S002	523
4	S003	506
5	S001	482

What this shows: S004, S005 and S002 are the main store-level stockout priorities.

7. Stockout Impact

Q19 — Revenue During Stockout Records

Which stores had stockouts, and how much revenue was recorded during those stockout records?

Store	Revenue During Stockout	Stockouts
S004	51,942,772.26 currency units	547
S002	48,491,757.02 currency units	523
S005	47,897,768.87 currency units	527
S003	47,175,660.93 currency units	506
S001	45,575,660.09 currency units	482

What this shows: S004 has both the highest stockout count and the highest revenue recorded during stockout records.

Q20 — Below-Average Fulfillment

Which stores have fulfillment rates below the company average?

Store	Average Fulfillment Rate
S002	98.12
S004	98.46
S005	100.25
S003	100.34

What this shows: S002 has the lowest average fulfillment rate among the stores returned by the query.

Q21 — Category Revenue During Stockout Records

Which categories generate the most revenue during stockout records?

Category	Revenue During Stockout	Stockouts
Furniture	51,564,040.97 currency units	524
Groceries	51,338,908.08 currency units	535
Toys	48,462,791.11 currency units	508
Electronics	46,553,132.68 currency units	508
Clothing	44,201,103.33 currency units	510

What this shows: Furniture has the highest revenue recorded during stockout records, followed by Groceries.

8. Stockout Demand Gap & Final Takeaways

Q22 — Demand Gap During Stockout Records

Do stores with more stockouts also have bigger demand gaps during those stockouts?

Store	Stockouts	Demand Gap During Stockouts
S004	547	7,225.39
S005	527	6,870.19
S003	506	6,677.72
S002	523	6,674.91
S001	482	6,149.35

What this shows: S004 has the highest stockout count and the highest demand gap during stockout records. This is an observed pattern, not proof that stockouts cause the demand gap.

Overall SQL Takeaways

- Revenue is broadly distributed across regions and stores, with East and S005 leading their groups.
- Stockouts occur across every region, store and category. Groceries have the highest category count, while S004 has the highest store count.
- Demand gaps are spread across the business. Toys have the highest category demand gap, while S005 has the highest store-level demand gap.
- S004, S005 and S002 stand out as the stores with the highest stockout counts.
- The SQL analysis helps identify where inventory availability and demand-fulfillment problems deserve further attention.